EDUCATIVE COMMENTARY ON ISI 2022 MATHEMATICS PAPERS

(Last Revised on 14/05/2025)

Introduction

The original motivation for writing the annual commentaries on ISI B.Stat. Entrance Tests is now waning. They began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continued more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests. But things are changing here too. In addition to the official key to the multiple choice questions in Paper 1, videos giving worked out solutions are available. As for the second paper, solutions for some of the problems are presented on the Artofsolvingproblems.

But even with both the justifications for the commentaries losing their strengths, I believe there is still some room to bring out the educative aspects of some of the problems and solutions. So, here is the commentary on ISI 2022.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, Third Edition, published by Universities Press, Hyderabad.

I am thankful to the persons who made the question papers available to me. Siddharth gave the alternate solutions to Q.3 and Q.24 in Paper 1.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2022 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Any positive real number x can be expressed as

$$x = a_n 2^n + a_{n-1}2^{n-1} + \ldots + a_1 2^1 + a_0 2^0 + a_{-1}2^{-1} + a_{-2}2^{-2} + \ldots$$

for $n \geq 0$, where each $a_i \in \{0, 1\}$. In the above-described expansion of 21.1875, the smallest positive integer k such that $a_{-k} \neq 0$ is

(A) 3 (B) 2 (C) 1 (D) 4

Answer and Comments: (A). The statement of the problem is needlessly elaborate. It simply gives the definition of what is popularly called the binary or dyadic expansion of a positive real number. It is analogous to the decimal expansion except that the base is not 10 but 2. As a result, the number x is expressed as a sum of powers of 2. The integral part of x involves only the powers with non-negative exponents. The fractional part requires powers with negative exponents. Instead of writing them as $2^{-1}, 2^{-2}, 2^{-3}$ etc. we resort to the more familiar expressions like $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$ etc. Moreover, since we are given the number x in its decimal expansion, we shall write these negative powers as $0.5, 0.25, 0.125$ etc. for easy comparison.

Now coming to the problem itself, we ignore the integral part of x and concentrate only on its fractional part, viz., 0.1875. It is smaller than 0.5. Hence the first digit of dyadic expansion is 0, i.e. $a_{-1} = 0$. Similarly, $a_{-2} = 0$ since 0.1875 is smaller than 0.25 too. However, it is bigger than 0.125 and so $a_3 \neq 0$. Therefore $k = 3$. ∎

An absolutely simple problem once a candidate deciphers the (unnecessarily) elaborate statement and realises that what is asked is the binary expansion of the fractional part of 0.1875. Perhaps the paper setters wanted to test this very ability of interpreting the given statement in terms of something familiar. The problem could have been made more interesting, had the entire dyadic expansion of 0.1875 been involved. To find it we subtract 0.125 from 0.1875 and expand the difference 0.0625 in terms of further negative powers of 2, viz. $\frac{1}{16}, \frac{1}{32}$ and so on. Luckily, 0.0625 equals $\frac{1}{16}$ and so the dyadic expansion of the fractional part stops at 0.0011. (Note that in the dyadic expansion the only coefficients are 0 and 1. So we do not have to worry the highest multiple of a particular power that is less than the earlier power.)

Q.2 Suppose for some $\theta \in [0, \frac{\pi}{2}]$, $\frac{\cos 3\theta}{\cos\theta} = \frac{1}{3}$. Then $(\cot 3\theta)\tan\theta$ equals

$$\text{(A) } \frac{1}{2} \quad \text{(B) } \frac{1}{3} \quad \text{(C) } \frac{1}{8} \quad \text{(D) } \frac{1}{7}$$

Answer and Comments: (D). There are well known identities which express $\cos 3\theta$ and $\sin 3\theta$ as cubic polynomials in $\cos\theta$ and $\sin\theta$ respectively. We can use the first one out of these and the data to get a quadratic in $\cos\theta$. Solving, we can get the value of $\cos\theta$ and hence that of $\sin\theta$. We can then use the second identity to get the numerical value of $\sin 3\theta$. From this the values $\tan\theta$ and thereafter those of $\tan 3\theta$ and $\cot 3\theta$ would follow. This would be a very straightforward way of getting the answer.

But there is a short cut. The expression, say E, to be evaluated is simply $\frac{\cos 3\theta}{\cos\theta}\frac{\sin\theta}{\sin 3\theta}$. The value of the first fraction is given as $\frac{1}{3}$. So, if we can determine the second factor, or its reciprocal $\frac{\sin 3\theta}{\sin\theta}$, we would be through.

The identity $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ gives

$$\frac{\sin 3\theta}{\sin\theta} = 3 - 4\sin^2\theta \tag{1}$$

Thus the task is reduced to finding the value of $\sin^2\theta$, *without* finding the value of $\sin\theta$. (Or else, this is hardly a short cut!). If we write $\cos 3\theta$ as $\cos(2\theta + \theta)$, we have

$$\begin{aligned} \cos 3\theta &= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta \\ &= (1 - 2\sin^2\theta)\cos\theta - 2\sin^2\theta\cos\theta \\ &= (1 - 4\sin^2\theta)\cos\theta \qquad (2) \end{aligned}$$

As we are given that $\cos 3\theta = \frac{1}{3}\cos\theta$, we get $\frac{1}{3} = 1 - 4\sin^2\theta$ and hence $4\sin^2\theta = 1 - \frac{1}{3} = \frac{2}{3}$, giving, finally, $\sin^2\theta = \frac{1}{6}$. Putting this into (1) gives $\frac{\sin 3\theta}{\sin\theta} = 3 - \frac{2}{3} = \frac{7}{3}$ and hence $E = \frac{1}{3} \times \frac{3}{7} = \frac{1}{7}$. ▮

A simple problem based on trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$. But, instead of applying these blindly, $\cos 3\theta$ has been found by writing it as $\cos(2\theta+\theta)$. So here is an instance where it is not the final formula but a step in its proof that shortens the work.

Q.3 The locus of points z in the complex plane satisfying $z^2 + |z|^2 = 0$ is

(A) A straight line
(B) a pair of straight lines
(C) a circle
(D) a parabola

Answer and Comments: (A). Every complex number $z = x + iy$ is uniquely determined by two real numbers, x and y. So, in theory, every problem about complex numbers can be reduced to a problem about their real and imaginary parts. This is not always a good idea, But locus problems are an exception, especially when we are asked merely to identify the nature of the locus, because till the HSC level only a few loci are studied and they are all in terms of the coordinates.

So, if $z = x + iy$, then the given equation translates into

$$x^2 - y^2 + 2ixy + x^2 + y^2 = 0 \tag{1}$$

As the real and imaginary parts must vanish separately, this is a pair of two equations in x and y, viz. (i) $2x^2 = 0$ and (ii) $xy = 0$. These two can hold simultaneously only when $x = 0$. So, this is the locus of z. It is a straight line, in fact, the y-axis. ∎

Another absolutely simple problem which can also be done by factorising of the L.H.S. as $(z + i|z|)(z - i|z|)$. Since $|z|$ is real, vanishing of either factor implies that z is purely imaginary and hence lies on the y-axis, which is a straight line.

Q.4 Amongst all polynomials $p(x) = c_0 + c_1x + c_2x^2 + \ldots + c_{10}x^{10}$ with real coefficients satisfying $|p(x)| \leq |x|$ for all $x \in [-1, 1]$, what is the maximum possible value of $(2c_0 + c_1)^{10}$?

(A) 4^{10} (B) 3^{10} (C) 2^{10} (D) 1

Answer and Comments: (D). The fact that all the four options are given as 10-th powers suggests that the real problem is to find the maximum possible value of $2c_0 + c_1$ subject to the constraint on $p(x)$, viz. $|p(x)| \leq |x|$ for all $x \in [-1, 1]$. Since $p(0) = c_0$, the requirement $|p(0)| \leq |0| = 0$ forces c_0 to vanish. Hence $p(x) = c_1x + c_2x^2 + \ldots + c_{10}x^{10}$. This can be factored as $xq(x)$ where $q(x) = c_1 + c_2x + \ldots + c_{10}x^9$. Then $|p(x)| = |x||q(x)|$. So the requirement $|p(x)| \leq |x|$ now gives $|x||q(x)| \leq |x|$ for all $x \in [-1, 1]$ and further, $|q(x)| \leq 1$ for $x \neq 0$ since $|x| > 0$ for $x \neq 0$. But since $q(x)$ is a continuous function of x, we must have $|q(0)| \leq 1$ too. This means $|c_1| \leq 1$. Hence the maximum possible value of $2c_0 + c_1$ among all polynomials of the given type is simply 1 and its 10-th power, viz. 1 is the answer. ∎

A simple but unusual problem. Getting $c_0 = 0$ is easy. A rigorous proof of $|c_1| \leq 1$ requires continuity of $q(x)$ at $x = 0$. Candidates who miss this are rewarded in terms of time. Ignorance is bliss in an MCQ.

Q.5 Let $\mathbb{Z}$ denote the set of all integers. Let $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be such that $f(x)f(y) = f(x+y) + f(x-y)$ for all $x, y \in \mathbb{Z}$. If $f(1) = 3$, then $f(7)$ equals

(A) 840 (B) 844 (C) 843 (D) 842

Answer and Comments: (C). The given relationship among the values assumed by f, viz.

$$f(x)f(y) = f(x+y) + f(x-y) \tag{1}$$

for all $x, y \in \mathbb{Z}$ is an example of what is called a functional equation. They are dealt with in Chapter 20. But because of the huge variation in the types of relations, there is not much of a coherent theory for for solving functional equations involving two variables, as there is, for example, for recurrence relations with only one variable (e.g. A.P.'s, the G.P.'s or the Fibonacci relation). So every problem requires methods tailor made for it. In the present case we notice that if we interchange x and y, only the last term in (1) changes and we get

$$f(x-y) = f(y-x) \tag{2}$$

for all $x, y \in \mathbb{Z}$. Since any integer can be expressed as a difference of two integers, we get that f is an even function.

We are further given that $f(1) = 3$. Putting $y = 1$ in (1), we have

$$3f(x) = f(x+1) + f(x-1) \tag{3}$$

for all $x \in \mathbb{Z}$. If we were given $f(1)$ as 2 instead of 3, then this equation would mean that $f(x)$ is an A.P., a most simple linear recurrence relation of order 2. Even as (3) stands, it is a linear recurrence relation of order 2. To solve it we need to know two initial values. In the present problem we are given, only one, viz. $f(1) = 3$. But, putting $x = 0$ into (3), and using that $f(-1) = f(1) = 3$, we get $f(0) = 2$.

Replacing x by $x - 1$, we can rewrite (3) as

$$f(x) = 3f(x-1) - f(x-2) \tag{4}$$

for all $x \in \mathbb{Z}$. For those who are familiar with solving linear recurrence relations with constant coefficients, the characteristic polynomial here is $\lambda^2 - 3\lambda + 1$. Its roots, viz. $\dfrac{3 \pm \sqrt{5}}{2}$ are the characteristic roots. Hence the general solution of (4) is

$$f(x) = A(\frac{3+\sqrt{5}}{2})^x + B(\frac{3-\sqrt{5}}{2})^x \tag{5}$$

where A and B are some constants to be determined from the initial conditions $f(0) = 2$ and $f(1) = 3$. Once this is done, $f(7)$ can be found by a mere substitution.

But since 7 is a small number, such an elaborate procedure can be avoided by simply calculating $f(2), f(3), f(4), f(5), f(6)$ and $f(7)$ one by one. So,

$$\begin{aligned}
f(2) &= 3f(1) - f(0) = 9 - 2 = 7 \\
f(3) &= 3f(2) - f(1) = 21 - 3 = 18 \\
f(4) &= 3f(3) - f(2) = 54 - 7 = 47 \\
f(5) &= 3f(4) - f(3) = 141 - 18 = 123 \\
f(6) &= 3f(5) - f(4) = 369 - 47 = 322
\end{aligned}$$

Hence, finally, $f(7) = 3f(6) - f(5) = 966 - 123 = 843$. ∎

The calculations above are like a passenger train from 1 to 7 which halts at every intermediate station 2,3,4,5 and 6. We could instead take an express train halting only at 3 and 5, assuming that we have already calculated $f(2)$ as 7. Note that (3) was obtained by putting $y = 1$ in (1). If, instead, we put $y = 2$, we get

$$7f(x) = f(x + 2) + f(x - 2) \tag{6}$$

Starting from $x = 1$, we get $21 = f(3) + f(-1) = f(3) + 3$. Hence $f(3) = 18$. So far, there is not much saving. But now put $x = 3$ into (6) to get $7f(3) = f(5) + f(1)$ which gives $f(5) = 7 \times 18 - 3 = 123$. Finally, knowing $f(5)$ and $f(3)$, we get $f(7)$ quickly by putting $x = 5$.

Of course, there is a super express train if we put $y = 3$ in (1) to get

$$18f(x) = f(x + 3) + f(x - 3) \tag{7}$$

Now the only intermediate stop needed is $f(4) = 47$.

A very good problem on functional equations where the functional relation involves two variables. Reduction to a functional equation with only one parameter itself involves a trick. The paper setters hardly expected the general solution in the form (5). Instead, repeated applications of the recurrence relation gives the answer. But this could have been tested equally well even with finding $f(3)$ or $f(4)$. Calculations beyond that are repetitious and prone to error, even if we take the express train above.

Overall, the question is more suited as a full length question in Paper 2.

Q.6 Let A and B be two 3×3 matrices such that $(A + B)^2 = A^2 + B^2$. Which of the following must be true?

(A) A and B are zero matrices.

(B) AB is the zero matrix.

(C) $(A - B)^2 = A^2 - B^2$.

(D) $(A - B)^2 = A^2 + B^2$

Answer and Comments: (D). As matrix multiplication is not commutative, the correct expansion of $(A + B)^2$ is $(A + B)(A + B) = A^2 + AB + BA + B^2$. So the hypothesis means that $AB + BA = O$, the zero matrix. Neither of the first two options would follow conclusively from this. The other two options deal with $(A-B)(A-B)$ which comes out to be $A^2 - AB - BA + B^2$. This equals $A^2 + B^2$ since $AB + BA = O$. ∎

Another trivial problem.

Q.7 Let $\{n_1, n_2, \ldots, n_{12}\}$ be a permutation of the numbers $1, 2, \ldots, 12$. The number of arrangements with

$$n_1 > n_2 > n_3 > n_4 > n_5 > n_6$$

and

$$n_6 < n_7 < n_8 < n_9 < n_{10} < n_{11} < n_{12}$$

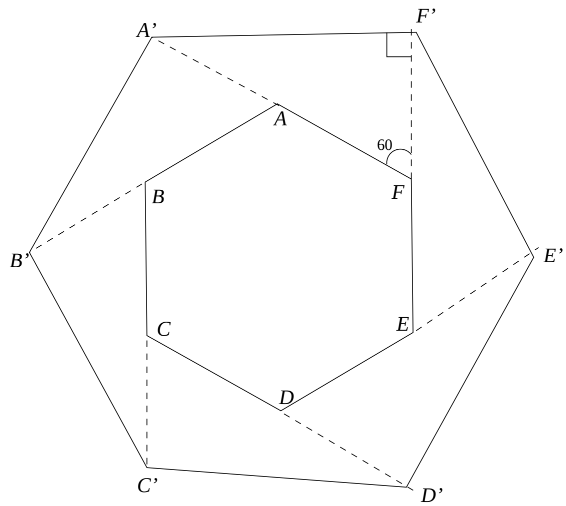
equals

(A) $\binom{12}{5}$ (B) $\binom{12}{6}$ (C) $\binom{11}{6}$ (D) $\frac{11!}{2}$

Answer and Comments: (C). Clearly n_6 is the smallest element of the set $\{1, 2, \ldots, 12\}$, viz. 1. The elements n_1, n_2, n_3, n_4 and n_5 can be any five of the remaining 11 elements. They can be chosen in $\binom{11}{5}$ ways. Once the choice is made, they must be ordered in a descending order. And after that the remaining six elements have to be ordered in a strictly increasing order to get n_7 to n_{12}. Summing up, the five elements chosen from the set $\{2, 3, \ldots, 12\}$ uniquely determine the permutation. Hence the number of permutations which satisfy the given requirements is $\binom{11}{5}$ which is the same as $\binom{11}{6}$. ∎

A simple problem once the idea that $n_6 = 1$ and that the choice of the subset $\{n_1, \ldots, n_5\}$ completely determines the permutation strikes.

Q.8 The sides of a regular hexagon $ABCDEF$ are extended by doubling them to form a bigger hexagon $A'B'C'D'E'F'$ as in the figure below.



Then the ratio of the area of the bigger to the smaller hexagon is

(A) $\sqrt{3}$ (B) 3 (C) $2\sqrt{3}$ (D) 4

Answer and Comments: (B). Being regular, both the two hexagons are similar to each other. Hence the ratio of their areas is the square of the ratio of their corresponding sides. It is tempting to think that this ratio is 2 (and hence that the ratio of the areas is 4) because the bigger hexagon is obtained by doubling the sides of the original hexagon. But the catch is that we are only extending the sides of the original hexagon. These extended segments are *not* the sides of the bigger hexagon. For example, in the figure above, $A'F$ is twice the side AF of the smaller hexagon. But it is not a side of the bigger hexagon. Instead, $A'F'$ is a side. Its length can be determined from the right angled triangle $A'F'F$ in which the angle at F is $60°$. Hence $A'F' = A'F \sin 60° = 2AF\frac{\sqrt{3}}{2} = \sqrt{3}AF$. The ratio of the corresponding sides being $\sqrt{3}$, that of the areas is 3. ∎

The paper setters have done well to give a diagram to expel the possible misconception that the sides of the bigger hexagon are double those of the smaller one. Calculating the ratio of the sides is very easy. The only catch is to realise that the ratio of the areas is the square of the ratio of the corresponding sides.

Q.9 In how many ways can we choose $a_1 < a_2 < a_3 < a_4$ from the set $\{1, 2, \ldots, 30\}$ such that a_1, a_2, a_3, a_4 are in arithmetic progression ?

(A) 135 (B) 145 (C) 155 (D) 165

Answer and Comments: (A). Let d be the common difference of the A.P. Then $a_4 - a_1 = 3d$ which is a multiple of 3. In order that both a_1 and a_4 are in the set $\{1, 2, \ldots, 30\}$, $3d$ can be at most 27. Hence the possible values of d are $1, 2, \ldots, 9$. If $d = 1$, then the possible values of the initial term a_1 are from 1 to 27. For $d = 2$, they are from 1 to 24. Continuing, the total number of desired progressions is $27 + 24 + 21 + 18 + 15 + 12 + 9 + 6 + 3$. This itself is the sum of an A.P. and equals $3(1 + 2 + \ldots + 9) = 3\frac{9\times 10}{2} = 135$. ∎

A simple counting problem. It could have been made a little more interesting by asking the probability that four distinct integers picked at random from the set $\{1, 2, \ldots, 30\}$ form an A.P.

Q.10 Suppose the numbers $71, 104$ and 159 leave the same remainder r when divided by a certain number $N > 1$. Then the value of $3N + 4$ is

(A) 53 (B) 48 (C) 37 (D) 23

Answer and Comments: (A). The hypothesis implies that there are some integers a, b, c such that

$$
\begin{aligned}
71 &= aN + r && (1)\\
104 &= bN + r && (2)\\
\text{and} \quad 159 &= cN + r && (3)
\end{aligned}
$$

The first two equations give $33 = (b - a)N$. Similarly, the second and the third equations give $55 = (c - b)N$. Hence N is a common divisor of 33 and 55. Since $N > 1$, we must have $N = 11$. The first equation along with the restriction on the remainder, viz. that $0 \leq r \leq 10$ gives $a = 6$ and $r = 5$. There is no point in trying the other two equations, since they will give the same value of r. Since $N = 11$ and $r = 5$, we have $3N + 4r = 33 + 20 = 53$. ∎

A very simple number theory problem. In fact, a disappointment on the backdrop of some of the past years when more challenging questions based on divisibility and the euclidean division algorithm were asked. (e.g. see Problem 13 of Paper 1 of last year.)

Q.11 If x, y are positive real numbers such that $3x + 4y < 72$, then the maximum possible value of $12xy(72 - 3x - 4y)$ is :

(A) 12240 (B) 13824 (C) 10056 (D) 8640

Answer and Comments: (B). Positive constant coefficients have little role in problems of maximisation. The paper-setters could as well have dropped 12 and asked to maximise $xy(72 - 3x - 4y)$. If nothing else, at least the options would not have looked so horrendous. So, perhaps there is some purpose in retaining it. And indeed there is, if we recognise that $12xy$ is the product of $3x$ and $4y$, about which something is given, viz. that their sum is less than 72.

So, if we let $u = 3x, v = 4y$ and $w = 72 - 3x - 4y$, then u, v, w are positive real numbers which add to 72 and we have to find the maximum value of their product uvw. By the A.M.-G.M. inequality, uvw is maximum when u, v, w are equal. This gives $u = v = w = 24$ and the maximum value of the product is $(24)^3$. The paper-setters have given one more reward. Even without calculating it fully, $(24)^3$ has 4 as its last digit. Out of the four options, only (B) has this property. So the right answer must be 13824. ∎

The problem amounts to maximising a function of two variables, viz. $12xy(72-3x-4y)$ subject to the inequality constraints $x > 0, y > 0$ and $3x + 4y < 72$. Doing this using calculus falls outside the scope of HSC. But with the A.M.-G.M. inequality, the problem becomes simple. The paper setters deserve to be commended for giving a veiled hint through the factor 12 which is otherwise redundant and further for giving only one option whose last digit matches that of $(24)^3$, thereby sparing the drudgery of calculating it.

Q.12 What is the minimum value of the function $|x-3|+|x+2|+|x+1|+|x|$ for real x?

(A) 3 (B) 5 (C) 6 (D) 8

Answer and Comments: (C). Call the function as $f(x)$. Then $f(x)$ is continuous everywhere and differential at all points except $-2, -1, 0$ and 3. Clearly, $f(x) \to \infty$ as $x \to \pm\infty$. Hence $f(x)$ has no maximum. Its minimum must occur at a critical point. If we sketch the graph of $f(x)$, we can tell its minimum value. But it is very time consuming to do so because the formula for $f(x)$ changes over the intervals $(-\infty, -2], [-2, -1], [-1, 0], [0, 3]$ and $[3, \infty)$. But on each of these intervals it is a linear function of x, i.e. a function of the form $Ax + B$ for some constants A and B (which are generally different for the five intervals). The maximum as well as the minimum of $f(x)$ over each such interval occurs at an end point. (If $A = 0$ for some interval, then $f(x)$ is constant over that interval and the extrema also occur at the end points.)

Summing up, even without drawing the graph of $f(x)$, we know that its minimum must occur at one of the points $-2, -1, 0$ and 3. By a direct calculation, $f(-2) = 2+1+5 = 8, f(-1) = 1+1+4 = 6, f(0) = 1+2+3 = 6$ and $f(3) = 3+4+5 = 12$. Hence the minimum occurs at both -1 and 0. (In fact, it occurs at every point in the interval $[-1, 0]$ since the function is constant over this interval.) The minimum value of f is $f(-1) = f(0) = 6$. ▮

The search for the minimum of a non-negative continuous function $f(x)$ on $I\!R$ narrows down to its critical points. These are the points where $f'(x)$ either does not exist or exists but vanishes. In the present problem, the latter possibility is taken care of by the constancy of $f(x)$ in a neighbourhood of such points. Once again, the MCQ format fails to distinguish whose who can give this justification from those who don't even realise that a justification is needed. So this question deserves a better place in the second paper.

Q.13 Consider a differentiable function $u : [0, 1] \longrightarrow I\!R$. Assume the function u satisfies

$$u(a) = \frac{1}{2r} \int_{a-r}^{a+r} u(x)dx, \text{for all } a \in (0, 1) \text{ and all } r < \min\{a, 1-a\}.$$

Which of the following statements must be true?

(A) u attains its maximum but not its minimum on the set $\{0, 1\}$.

(B) u attains its minimum but not maximum on the set $\{0, 1\}$.

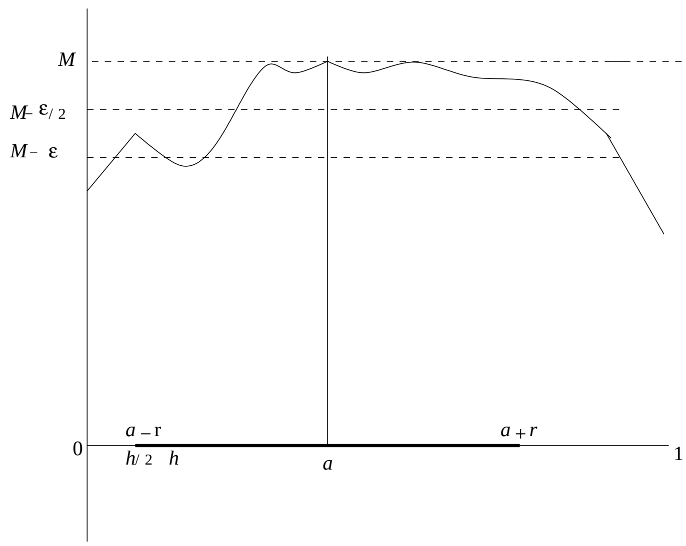
(C) If u attains either its maximum or its minimum on the set $\{0, 1\}$, then it must be a constant.

(D) u attains both its maximum and its minimum on the set $\{0, 1\}$.

Answer and Comments: (D). The property which u is given to satisfy may appear clumsy. The expression $\dfrac{1}{2r}\displaystyle\int_{a-r}^{a+r} u(x)dx$ is called the average value of $u(x)$ over the interval $[a-r, a+r]$. More generally, the **average** or the **mean** value of a continuous real valued function $f(x)$ over an interval $[a, b]$ is defined as $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)dx$. It is the continuous analogue of the average value of a real valued function defined on a finite set and shares some common properties. For example, it always lies between the minimum and the maximum of the function over that interval. The average value of a real valued function on a finite set may not always be attained. (In a class of students, there may be no student whose height is exactly the mean height for that class.) In sharp contrast, a continuous function $f(x)$ always attains its average. This follows by applying the Lagrange's Mean Value theorem to any antiderivative F of f and justifies the name mean value.

Coming to the problem, we are given that the average value of u over any interval contained in $(0, 1)$ occurs at the midpoint of that interval. This is a very strong hypothesis and from it we can show that u is a linear function, i.e. a function of the form $u(x) = Ax + B$ for some constants A and B. Depending on the sign of A, u attains its maximum on $[0, 1]$ at one of the end points and minimum at the other. That is, (D) holds. But the proof of linearity of u is non-trivial. (The converse is trivial, because if $u(x) = Ax + B$, then $\displaystyle\int_a^b u(x)dx = \frac{A}{2}(b^2 - a^2) + B(b-a) = (b-a)u(\frac{a+b}{2})$.)

As our goal is merely to prove (D), we can give a simpler argument based merely on continuity of u. u has a maximum, say M, at some point $a \in [0, 1]$. If either $u(0)$ or $u(1)$ equals M, we are done. So assume that both $u(0) < M$ and $u(1) < M$. Choose $\epsilon > 0$ such that $u(0) < M - \epsilon$ and $u(1) < M - \epsilon$. By continuity of u at 0 and 1, there exists some $h > 0$ such that $u(x) < M - \epsilon/2$ for all $x \in [0, h]$ and also for all $x \in [1-h, 1]$. Now let $r = \min\{a - \frac{h}{2}, 1 - \frac{h}{2} - a\}$. Then $[a-r, a+r] \subset (0,1)$. The situation is shown in the figure below where it is assumed, without loss of generality, that a lies in the left half of $[0,1]$ and hence $a - r = \frac{h}{2}$. The interval $[a-r, a+r]$ is shown in thick.



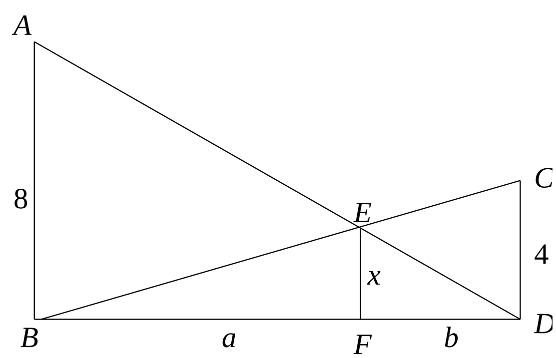
Now let $I = \displaystyle\int_{a-r}^{a+r} u(x)dx$. Then $u(x) \leq M$ for all $x \in [a-r, a+r]$ and hence $I \leq 2rM$. We split the interval of integration at the point h. The first part is of length $\frac{h}{2}$ while the second part is of length $2r - \frac{h}{2}$. On the first subinterval, $u(x) \leq M - \frac{\epsilon}{2}$. Hence $\displaystyle\int_{h/2}^{h} u(x)dx \leq \frac{h}{2}(M - \frac{\epsilon}{2})$. The integral of $u(x)$ over the second interval is at most $(2r - \frac{h}{2})M$. Adding,

$$
\begin{aligned}
I &\leq \frac{h}{2}(M - \frac{\epsilon}{2}) + (2r - \frac{h}{2})M \\
&= 2rM - \frac{h\epsilon}{4} \qquad (1)
\end{aligned}
$$

So, the average value of u over the interval $[a-r, a+r]$ is $M - \frac{h\epsilon}{8r}$. This is less than M which equals $u(a)$ by assumption. Thus we get a contradiction. This proves that u attains its maximum at an end point of $[0,1]$. By a similar reasoning it attains its minimum at an end point. Hence (D) is true. ∎

The essential idea behind the argument above is merely that if M is an upper bound on a continuous integrand over an interval of length L, then the integral is at most ML. But even if the upper bound on the integrand is strict at one point then by continuity it is so over a subinterval of positive length and therefore the integral is less than ML. Using essentially the same argument, we can prove a slightly stronger result that $u(x)$ is either monotonically increasing or decreasing. This is, of course, a far cry from proving its linearity. Note that differentiability of u has never been used. The purpose of giving it in the problem is not clear. Perhaps with it the linearity of u can be proved. At any rate, this is an excellent problem on theoretical calculus and more suited for Paper 2.

Q.14 A straight road has walls on both sides of heights 8 feet and 4 feet respectively. Two ladders are placed from the top of one wall to the foot of the other as in the figure below. What is the height (in feet) of the maximum clearance x below the ladders?



(A) 3 (B) $2\sqrt{2}$ (C) $\frac{8}{3}$ (D) 1

Answer and Comments: (C). It might appear at first sight that there is no unique answer since the width of the road, i.e. the distance between the two walls, is not specified. So let us try to solve the problem with some unknown parameters. Call the walls as AB and CD. Let the ladders meet at E and EF be the vertical clearance. Denote BF and FD by a and b respectively. From the similar triangles BEF and BCD, we get

$$\frac{x}{4} = \frac{a}{a+b} \tag{1}$$

Similarly, the similar triangles EFD and ABD give

$$\frac{x}{8} = \frac{b}{a+b} \tag{2}$$

Adding these two equations,

$$\frac{x}{4} + \frac{x}{8} = 1 \tag{3}$$

which simplifies to $12x = 32$ and gives $x = \frac{8}{3}$. ∎

An absolutely simple problem requiring nothing more than elementary properties of similar triangles.

Q.15 Let $y = x + c_1$, $y = x + c_2$ be two tangents to the ellipse $x^2 + 4y^2 = 1$. What is the value of $|c_1 - c_2|$?

(A) $\sqrt{2}$ (B) $\sqrt{5}$ (C) $\frac{\sqrt{5}}{2}$ (D) 1

.

Answer and Comments: (B). We could begin by casting the equation of the ellipse in the standard form and then tapping on the right tools such as the formulas for equations of tangents or representing the ellipse in a parametric form. But it is often better to resort to general principles and proceed directly. The slope of the tangent at a point (x_0, y_0) on any curve whose equation is given in terms of x and y is the value of $\frac{dy}{dx}$ at that point. Here y is not given as an explicit function of x. But we can apply implicit differentiation and get

$$2x + 8y\frac{dy}{dx} = 0 \tag{1}$$

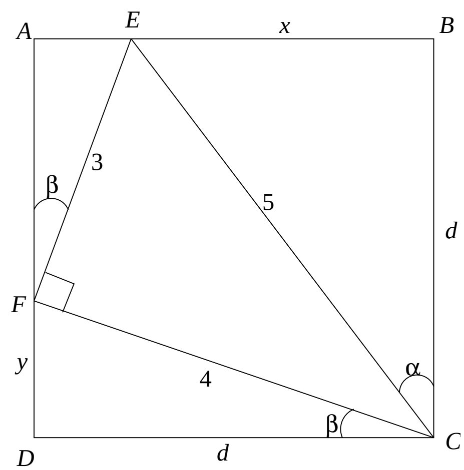
If (x_1, y_1) is the point of contact of the tangent $y = x + c_1$, then $\frac{dy}{dx}$ at this point is 1 and we get $x = -4y$. Solving this simultaneously with $x^2 + 4y^2 = 1$, we have $20y^2 = 1$. Hence $y = \pm\frac{1}{2\sqrt{5}}$. Correspondingly, $x = \mp\frac{2}{\sqrt{5}}$. So, the points of contact of the two tangents are $(\frac{-2}{\sqrt{5}}, \frac{1}{2\sqrt{5}})$ and $(\frac{2}{\sqrt{5}}, -\frac{1}{2\sqrt{5}})$. Call the first one as (x_1, y_1) and the second one as (x_2, y_2). Then the equation of the tangent at (x_1, y_1) is

$$y - \frac{1}{2\sqrt{5}} = x + \frac{2}{\sqrt{5}} \tag{2}$$

This gives $c_1 = \frac{5}{2\sqrt{5}}$. By a similar calculation, $c_2 = -\frac{5}{2\sqrt{5}}$. Therefore $|c_1 - c_2| = \frac{5}{\sqrt{5}} = \sqrt{5}$. ∎

A simple problem on conics which can be solved more easily through the general principles. In fact, the knowledge that the given curve is an ellipse is never needed.

Q.16 In the figure below, $ABCD$ is a square and $\triangle CEF$ is a triangle with given sides inscribed as in the figure. Find the length of BE.



(A) $\frac{13}{\sqrt{17}}$ (B) $\frac{14}{\sqrt{17}}$ (C) $\frac{15}{\sqrt{17}}$ (D) $\frac{16}{\sqrt{17}}$

Answer and Comments: (A). Whenever a problem asks to find the numerical value of something, the standard procedure is to denote the quantity by an unknown (often x), get an equation for x and solve it. But usually things are not so simple. Often we have to introduce some other auxiliary unknowns, y, z etc. and get a system of equations in all these unknowns. We solve it even though we want the value of only one of the unknowns.

Art lies in choosing these auxiliary variables. In the present problem, we can begin by calling BE as x. We can take the side of the square as d and the distance FD as y. The three right-angled triangles with hypotenuses along the sides of the given triangle CEF will give a system of three equations in the three unknowns x, y and d. We can then solve it. This will be a mechanical way of attempting the solution. And its success will depend upon how easy it is to solve this system. In the present problem, the system is algebraic but non-linear and *ad-hoc* methods will be needed to solve it.

A more imaginative way is to use some elementary trigonometry. Instead of using the Pythagoras theorem, we can introduce suitable angles and get a system in terms of these angles. So we let $\alpha = \angle BCE$. Then we have

$$x = 5\sin\alpha \tag{1}$$

If we can get hold of some trigonometric equation for α, we can get $\sin\alpha$ and thence x from it. But there is no such obvious equation. So we introduce an auxiliary variable. Specifically, we let $\beta = \angle DCF$. Equating the two expressions for the side d, we get

$$5\cos\alpha = 4\cos\beta \tag{2}$$

As there are two unknowns, we need one more equation relating α and β. It is tempting to get this by applying the Pythagoras theorem to $\triangle FAB$, since we have $AE = d - 5\sin\alpha = 5(\cos\alpha - \sin\alpha)$ and $FA = d - 4\sin\beta = 4(\cos\beta - \sin\beta)$. But squaring and adding will give a complicated equation in α and β and then this method would be as laborious as the algebraic one.

There is an ingenious way out if we observe that the triangle EFC is right angled at F. Hence $\angle AFB$ also equals β. Therefore, $AF = 3\cos\beta$. Adding this to FD we get d and hence a trigonometric equation for β, viz.

$$3\cos\beta + 4\sin\beta = d = 4\cos\beta \tag{3}$$

This gives $4\sin\beta = \cos\beta$. Hence $\tan\beta = \frac{1}{4}$, whence $\sin\beta = \frac{1}{\sqrt{17}}$ and $\cos\beta = \frac{4}{\sqrt{17}}$. From (2), we now have

$$5\cos\alpha = \frac{16}{\sqrt{17}} \tag{4}$$

Hence

$$x = 5\sin\alpha = \sqrt{25 - 25\cos^2\alpha}$$
$$= \sqrt{25 - \frac{256}{17}}$$
$$= \sqrt{\frac{425-256}{17}} = \sqrt{\frac{169}{17}} \qquad (5)$$

which comes to $\frac{13}{\sqrt{17}}$.

The trigonometry involved goes little beyond definitions. So it can be bypassed using similarity of the triangles FDC and EAF, which gives $AF : d = 3 : 4$ and further $y = d/4$ and $AE = 3d/16$. Applying Pythagoras theorem to $\triangle FDC$ gies $d = \frac{16}{\sqrt{17}}$. Hence $BE = d - 3d/16 = 13d/16 = \frac{13}{\sqrt{17}}$. ∎

An excellent problem testing the ability to select the right variables taking advantage of the specifics of the data. In the present problem $\triangle EFC$ is not just some triangle, but a right angled one and this led to a trigonometric equation in β.

Q.17 Let p and q be two non-zero polynomials such that the degree of p is less than or equal to the degree of q and $p(a)q(a) = 0$ for $a = 0, 1, 2, \ldots, 10$. Which of the following must be true?

(A) degree of $q \neq 10$

(B) degree of $p \neq 10$

(C) degree of $q \neq 5$

(D) degree of $p \neq 5$

Answer and Comments: (C). Let m and n be the degrees of p and q respectively. Then the degree of the product pq is $m + n$. We are given that pq has at least 11 distinct roots, viz. $0, 1, 2, \ldots, 10$. Hence the degree of pq, i.e. $m + n$ is at least 11. We are further given that $m \leq n$. So, the larger summand, viz. n is at least 6. Hence (C) is necessarily true. The falsity of the other options can be proved with easy counterexamples. But that is hardly necessary because of the stipulation that exactly one of the four given options is true. ∎

A shockingly simple problem requiring only that the degree of a product of polynomials is the sum of their degrees and that the number of roots of a polynomial cannot exceed its degree.

Q.18 For $n \in I\!N$, let a_n be defined by

$$a_n = \int_0^n \frac{1}{1+nx^2} dx.$$

Then $\lim\limits_{n\to\infty} a_n$

(A) equals 0 (B) equals $\frac{\pi}{4}$
(C) equals $\frac{\pi}{2}$ (D) does not exist.

Answer and Comments: (A). The integral can be evaluated explicitly with the substitution $y = \sqrt{n}x$. Then $dx = \frac{1}{\sqrt{n}}dy$ and we get

$$\begin{aligned} a_n &= \frac{1}{\sqrt{n}} \int_0^{n\sqrt{n}} \frac{1}{1+y^2} dy \\ &= \frac{1}{\sqrt{n}} \tan^{-1} y \Big|_0^{n\sqrt{n}} \\ &= \frac{\tan^{-1}(n\sqrt{n})}{\sqrt{n}} \end{aligned} \tag{1}$$

As $n \to \infty$, $n\sqrt{n} \to \infty$ and so the numerator tends to $\frac{\pi}{2}$. But the denominator tends to ∞. Hence the ratio tends to 0. So, $\lim\limits_{n\to\infty} a_n = 0$.
∎

A very simple problem since the integral can be evaluated explicitly. Limits of integrals with an integer parameter n are interesting when the integral cannot be evaluated explicitly but can only be estimated. See, for example, Exercise (18.18).

Q.19 A 3×3 magic square is a 3×3 rectangular array of positive integers such that the sum of the three numbers in any row, any column or any of the two major diagonals is the same. For the following incomplete magic square

27	36	
31		

the column sum is

(A) 90 (B) 96 (C) 94 (D) 99

Answer and Comments: (B). Looks like a typical number puzzle in the recreational section of a newspaper. Anyway the problem is purely mathematical. There are 9 entries of which 3 are given. The remaining 6 can be found by writing and solving a system of equations. As the only operations involved are addition and subtraction, the system is linear. But instead of writing it elaborately, we can solve parts of it one-by-one. For example, since the first row and the first column both add to the same number, if we call the last entry in the first row as x then the middle entry in the first column must be $x + 5$. Similarly, equality of $27 + 36 + x$ with the sum of the entries in the non-principal diagonal gives the central entry of the square as 32 and we get a picture below

27	36	x
x+5	32	z
31	u	v

where z, u, v are the remaining three unknowns. Equating the first two row sums determines z as 26. Similarly equality of the first two column sums determines u as $x - 5$. Finally, since both the diagonals have equal sums we get $v = x + 4$ and we get the more filled picture

27	36	x
x+5	32	26
31	x−5	x+4

where now there is only one unknown, viz. x. To find its value, there is no point in using equalities which are already used. But if we equate the first row sum with the third, we get $x + 63 = 2x + 30$ which gives $x = 33$. All the sums now are 96 each. ∎

A conceptually simple problem which is best done mentally on a piece of paper. The integrality of the entries was never used. The most general 3×3 magic square has 9 unknown entries which are best denoted as the entries a_{ij} of a 3×3 matrix. There are in all 8 sums (three row sums, three column sums and two diagonal sums) and their mutual equality gives a system of 7 homogeneous linear equations in the 9 unknowns, viz. the a_{ij}'s. It would thus appear superficially that this system has $9 - 7$, i.e. 2 degrees of freedom. In simpler terms, this means that any two entries of a 3×3 magic square determine it uniquely. But that is not correct. There is some redundancy in these 7 linear equations. For example, equality of each of the three row sums along with their equality with the first two column sums automatically implies that the third column sum also equals this common sum. The corresponding equation gives no new information. That reduces the number of 'really different' equations by 1 and increases the degrees of freedom from 2 to 3. So, 3 entries are needed to determine a magic square of order 3 uniquely. That is why in the present problem, three entries of the magic square are given. (The loose terminology of 'really different' equations and degrees of freedom can be made precise with the concept of the rank of a system of linear equations. But that is beyond the HSC level.)

Q.20 The number of positive integers n less than or equal to 22 such that 7 divides $n^5 + 4n^4 + 3n^3 + 2022$ is

(A) 7 (B) 8 (C) 9 (D) 10

Answer and Comments: (D). The number 2022 is just a showpiece, catering to the modern trend of giving at least one problem whose statement involves (often only superficially) the number of the calendar year in which a particular test is given. When divided by 7, 2022 leaves the remainder 6. So, the requirement about n is equivalent to saying that $n^5 + 4n^4 + 3n^3$ leaves a remainder of 1 when divided by 7. In a somewhat sophisticated language (given in Comment No. 15 of Chapter 4), $n^5 + 4n^4 + 3n^3$ is congruent to 1 modulo 7. Or, in other words, the residue class of $n^5 + 4n^4 + 3n^3$ modulo 7 is 1.

Using elementary properties of residue classes, this can be further simplified by factoring $n^5 + 4n^4 + 3n^3$ as $n^3(n^2 + 4n + 3)$ and further as $n^3(n + 1)(n + 3)$. The residue class modulo 7 of a sum and of a product are simply the sum and the product of the respective classes. So, if we denote $n^3(n + 1)(n + 3)$ by a_n, then the residue class of a_1 modulo 7 is $1^3 \times 2 \times 4 = 8 = 1$ since modulo 7, 8 is congruent to 1. Similarly, for $n = 2$, the residue of a_2 modulo 7 is $8 \times 3 \times 5 = 15 = 1$. For $n = 3$, a_3 has the residue class of $27 \times 4 \times 6$ and hence that of $(-1)4(-1) = 4$. The next case, viz. a_4 is much simpler, because when $n = 4$, $n + 3 = 7$ and so $n^3(n + 1)(n + 3)$ is divisible by 7, whence modulo 7 it is congruent to 0 and not to 1. For a similar reason, a_6 and a_7 are congruent to 0 modulo 7. As for a_5, its congruence class modulo 7 is that of $125 \times 6 \times 8$ and hence of $6 \times 6 \times 1 = 1$.

Summing up, out of a_1 to a_7 only three, viz. a_1, a_2 and a_5 fit the table. It would be torturous to calculate the residue classes modulo 7 of a_n for all n from 8 to 22. Fortunately, that is not necessary. Note that a_{n+7} has the same residue class modulo 7 as a_n because $n + 7$ is congruent to n modulo 7. So once we determine for how many values of n between 1 to 7, n divides $n^5 + 4n^4 + 3n^3$, the number would be the same for the next set of 7 integers, viz. from 8 to 14, and then again the next one, viz. the integers from 15 to 21. Only the case $n = 2$. But again, a_{22} is congruent to a_1 which is congruent to 1 modulo 7. So, in all the number of integers n from 1 to 22 for which 7 divides

$n^5+4n^4+3n^3$ is $3+3+3+1=10$. ∎

A good number theory problem based on residue arithmetic. The book keeping work could have been avoided. The essential idea of the problem is merely that the residue class of n modulo 7 determines that of $n^5+4n^4+3n^3$. And this could have been tested by asking to identify which of some given integers has the desired property.

Q.21 In a class of 45 students, three students can write well using either hand. The number of students who can write well only with their right is 24 more than the number of those who write well only with their left hand. Then the number of students who can write well with their right hand is

(A) 33 (B) 36 (C) 39 (D) 41

Answer and Comments: (B). A typical problem on the cardinalities of various subsets of a finite set. Let S, R, L and B be, respectively, the sets of all students, those who can write well only with their right hand, those who can write well only with their left hand, and those who can write well with either hand. Let s, r, l and b denote their respective cardinaities. Then S is the union of the three mutually disjoint subsets R, L, B. Hence $s = r + l + b$. We are given that $s = 45$ and $b = 3$. That gives $r + l = 45 - 3 = 42$. It is also given that $r - l = 24$. Hence $2r = \frac{42+24}{2} = 33$. The question asks for $r + b$ which equals $33 + 3 = 36$. ∎

Another shockingly simple problem. With only two subsets involved, there is not much point in drawing a Venn diagram either. It is not given explicitly that every student can write with at least one hand. This is left to common sense.

Q.22 Let $\mathbb{1}, \omega, \omega^2$ be the cube roots of unity. Then the product

$$(1-\omega+\omega^2)(1-\omega^2+\omega^{2^2})(1-\omega^{2^2}+\omega^{2^3})\ldots(1-\omega^{2^9}+\omega^{2^{10}})$$

is equal to

(A) 2^{10} (B) 3^{10} (C) $2^{10}\omega$ (D) $3^{10}\omega^2$

Answer and Comments: (A). All powers of ω are roots of unity. They recur in cycles of length 3. Thus $\omega^4 = \omega, \omega^5 = \omega^2$ etc. If $\alpha = \omega^k$, then $\alpha = 1$ if and only if k is a multiple of 3. Otherwise it is a complex cube root of unity and satisfies $1 + \alpha + \alpha^2 = 1$ and hence $1 - \alpha + \alpha^2 = -2\alpha$.

The r-th factor of the product, say P, is $1 - \omega^{2^r - 1} + \omega^{2^r}$. This is of the form $1 - \alpha + \alpha^2$ where $\alpha = \omega^{2^{r-1}}$. Since 2^{r-1} is not a multiple of 3, we get that the r-th factor equals $-2\alpha = -2\omega^{2^{r-1}}$. Therefore we have

$$\begin{aligned} P &= (-2)^{10} \prod_{r=1}^{10} (\omega^{2^{r-1}})^2 \\ &= 2^{10} \prod_{r=1}^{10} \omega^{2^r} \\ &= 2^{10} \omega^k \qquad (1) \end{aligned}$$

where

$$\begin{aligned} k &= \sum_{r=1}^{10} 2^r \\ &= \frac{2(2^{10} - 1}{2 - 1} \qquad (2) \end{aligned}$$

using the formula for the sum of a G.P. with common ratio 2. Since $2^{10} - 1 = 1023$ is divisible by 3, k is a multiple of 3 and therefore $\omega^k = 1$. Hence the product P equals 2^{10}. ∎

A good problem involving properties of ω. Such problems have become routine because many problems involving powers of ω have been asked in so many examinations. It would have been better to replace ω with a fifth or sixth root of unity. Then the same principles would apply as for ω but the calculations would demand some fresh work.

Q.23 The function $x^2 \log_e x$ in the interval $(0, 2)$ has:

(A) exactly one point of local maximum and no points of local minimum.

(B) exactly one point of local minimum and no points of local maximum.

(C) points of local maximum and as well as points of local minimum.

(D) neither a point of local maximum nor a point of local minimum.

Answer and Comments: (B). Let $f(x) = x^2 \ln x$. Then f is undefined at 0. But on the interval $(0, 2)$ it is defined and differentiable everywhere. Further,

$$f'(x) = 2x \ln x + x = x(2 \ln x + 1) \tag{1}$$

which vanishes when $\ln x = -\frac{1}{2}$, i.e. when $x = e^{-1/2} = \frac{1}{\sqrt{e}}$. Further, $f'(x) > 0$ for $x > \frac{1}{\sqrt{e}}$ and $f'(x) < 0$ for $x < \frac{1}{\sqrt{e}}$. Hence f is decreasing on $(0, \frac{1}{\sqrt{e}})$ and increasing on $(\frac{1}{\sqrt{e}}, 2)$. Therefore $\frac{1}{\sqrt{e}}$ is a point of local minimum. As there are no other critical points, the correct answer is (B). ∎

A straightforward problem since the end points 0 and 2 are not included and the question deals with only local extrema. If 0, were included in the domain, then as $x \to 0^+$, $\ln x \to -\infty$, but because of the factor x^2, the product $x^2 \ln x$ tends to 0.

Q.24 The number of triples (a, b, c) of positive integers satisfying the equation

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 + \frac{2}{abc}$$

and such that $a < b < c$ equals:

(A) 3 (B) 2 (C) 1 (D) 0

Answer and Comments: (C). If a, b, c were real numbers, then a single equation in three variables will generally have infinitely many solutions. But the stipulation that a, b, c are positive integers puts a strong restriction on the solutions. For example the equation $xy = 10$ represents a hyperbola with infinitely many points on it. But if x, y can take only positive integers as values then (x, y) can only be $(10, 1), (1, 10), (2, 5)$ and $(5, 2)$. Similarly, $x + y + z = 1$ is an infinite plane, but if x, y, z are non-negative integers, then the only possible solutions are $(1,0,0),(0,1,0)$ and $(0,0,1)$.

So, somewhere in the solution, we will need the restriction that a, b, c are positive e integers. But before invoking it, let us rewrite the given equation as

$$ab + bc + ca = abc + 2 \tag{1}$$

To recast this further we note the expansion

$$(1-a)(1-b)(1-c) = 1 - (a+b+c) + ab + bc + ca - abc \tag{2}$$

Adding 1 to both the sides of (1) we can rewrite it as

$$1 + ab + bc + ca - abc = 3 \tag{3}$$

which, in view of (2) becomes

$$(1-a)(1-b)(1-c) = 3 - (a+b+c) = (1-a) + (1-b) + (1-c) \tag{4}$$

or equivalently,

$$(a-1)(b-1)(c-1) = (a-1) + (b-1) + (c-1) \tag{5}$$

We have not yet solved the equation. We have merely recast it. But now we are close to the solution. Let us call $a-1, b-1, c-1$ as x, y, z respectively. Then x, y, z are non-negative integers. Moreover, $a < b < c$ is equivalent to $x < y < z$. Thus the problem is now reduced to finding the number of triples (x, y, z) of non-negative integers satisfying the equation

$$xyz = x + y + z \tag{6}$$

and such that $x < y < z$. This is a much more manageable task. Note that x cannot be 0. For it it is, then by (6), $x+y+z$ would be 0 forcing $x = y = z = 0$. But then the inequality $x < y < z$ fails. So x and therefore y and z are positive integers.

$(1,2,3)$ is an obvious solution of (6). We claim that it is the only solution. In any solution (x, y, z), we have $1 \leq x < y < z$. Hence $x + y + z < 3z$. That means $xyz < 3z$ which implies $xy < 3$ as $z > 0$.

As x, y are integers, this means $xy \leq 2$. But the only way this can happen with $1 \leq x < y$ is if $x = 1$ and $y = 2$.

So, there is only one solution of (6) and hence of the original equation (1) too. (The actual solution is $a = 2, b = 3, c = 4$.)∎

A tricky problem, where the success depends upon the conversion of (1) to (6). Last year Q.15 was also a problem where the requirement that $\max\{a, b, c, d\} = a^2 + b^2 + c^2 + d^2$ on the variables $a, b, c, d \in [0, 1]$ reduced the solution set from an infinite to a finite one. And that problem was tricky too.

A more methodical solution is also possible. As a, b, c are positive integers with $a < b < c$, they cannot be very large becuase in that case their reciprocals will be so small that they will add to something less than 1. That shows that the number of solutions is finite and we can find them by giving possible values to a.

For $a = 1$ the equation reduces to $\frac{1}{b} + \frac{1}{c} = \frac{2}{bc}$, which simplifies to $b + c = 2$. But this is impossible since $b \geq 2$ and $c \geq 3$. So there are no solutions with $a = 1$.

Suppose $a = 2$. Then the equation reduces to $\frac{1}{b} + \frac{1}{c} = \frac{1}{2} + \frac{1}{bc}$ which simplifies to $2(b+c) = bc+2$. This can be recast as $2(b-1)(c-1) = bc$. So $c - 1$ divides bc. But c and $c - 1$ are relatively prime. So we must have that $c - 1$ divides b. But $c - 1 \geq b$. This is possible only when $c - 1 = b$. Then the equation reduces to $2b - 2 = b + 1$ which gives $b = 3$ and hence $c = 4$. Thus $(2, 3, 4)$ is a solution.

For $a \geq 3$, we must have $b \geq 4$ and $c \geq 5$. But then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{60} < 1$. Hence the given equation cannot hold as the second term on the R.H.S. is positive.

Summing up, $(a, b, c) = (2, 3, 4)$ is the only solution. ∎

Q.25 An urn contains 30 balls out of which one is special. If 6 of these balls are taken out at random, what is the probability that the special ball is chosen?

(A) $\frac{1}{30}$ (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{1}{15}$

Answer and Comments: (C). It is intuitively obvious that the answer is $\frac{1}{5}$ because to be special among 30 is five times as remarkable as being special among 6. For a rigorous solution, note that it is easier to find the complementary probability, i.e. the probability that the special ball is *not* in the draw. For this to happen, all the 6 balls picked must be from the 29 balls that are not special. Hence the complementary probability is $\frac{\binom{29}{6}}{\binom{30}{6}}$. On expansion, this ratio equals $\dfrac{29 \times 28 \times 27 \times 26 \times 25 \times 24/6!}{30 \times 29 \times 28 \times 27 \times 26 \times 25/6!} = \frac{24}{30} = \frac{4}{5}$. Hence the desired probability is $1 - \frac{4}{5} = \frac{1}{5}$. ∎

A simple problem once the idea of complementary probability strikes.

Q.26 A triangle has sides of lengths $\sqrt{5}, 2\sqrt{2}, \sqrt{3}$ units. Then the radius of its inscribed circle is

(A) $\frac{\sqrt{5}+\sqrt{3}+2\sqrt{2}}{2}$ (B) $\frac{\sqrt{5}+\sqrt{3}+2\sqrt{2}}{3}$
(C) $\sqrt{5}+\sqrt{3}+2\sqrt{2}$ (D) $\frac{\sqrt{5}+\sqrt{3}-2\sqrt{2}}{2}$

Answer and Comments: (D). The radius of the incircle of a triangle is called the inradius of the triangle and generally denoted by r. There are several formulas for r. In the present problem, as we are given the three sides, say a, b, c, a formula in terms of the sides will be most directly applicable. One such formula is

$$r = \frac{S}{s} \tag{1}$$

where S is the area of the triangle and s is its semi-perimeter, defined as

$$s = \frac{a+b+c}{2} \tag{2}$$

We also need a formula for S in terms of a, b, c. There is a well-known formula, Heron's formula for this.

$$S = \sqrt{s(s-a)(s-b)(s-c)} \tag{3}$$

It is now merely a matter of substituting $a = \sqrt{5}, b = \sqrt{3}, c = 2\sqrt{2}$ into (2) and (3) and then get the numerical value of r from (1). The catch is that when we write (3) directly in terms of a, b, c, it becomes

$$S = \frac{1}{4}\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)} \tag{4}$$

As a, b, c are surds, the resulting expression for s after the substitution will be very complicated.

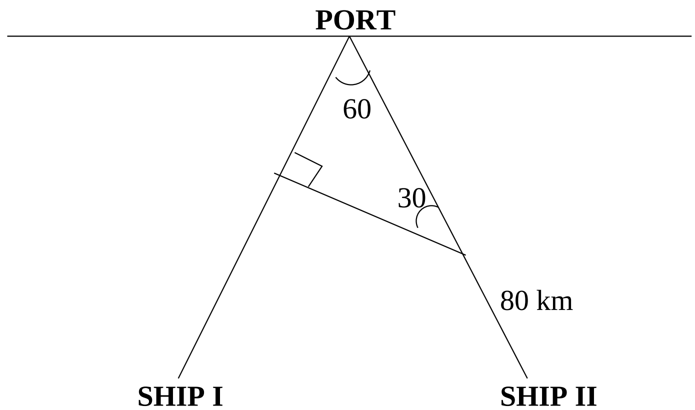
But there is a way out if we rewrite $2\sqrt{2}$ as $\sqrt{8}$. For, now the three sides are $\sqrt{5}, \sqrt{3}$ and $\sqrt{8}$. Since $5 + 3 = 8$, we see that the triangle is right angled, with $\sqrt{5}$ and $\sqrt{3}$ as its shorter sides. So, its area is simply $\frac{1}{2}\sqrt{15}$. So, instead of applying (3) we get from (1) and (2)

$$\begin{aligned} r &= \frac{\sqrt{15}}{\sqrt{5}+\sqrt{3}+2\sqrt{2}} \\ &= \frac{\sqrt{15}(\sqrt{5}+\sqrt{3}-2\sqrt{2})}{(\sqrt{5}+\sqrt{3})^2-8} \\ &= \frac{\sqrt{15}(\sqrt{5}+\sqrt{3}-2\sqrt{2})}{2\sqrt{15}} \\ &= \frac{\sqrt{5}+\sqrt{3}-2\sqrt{2}}{2} \end{aligned} \tag{5}$$

which is (D). ∎

Like Q.16, in this problem too, the specifics of the numerical data provides a short cut. Curiously, in both the problems, the key was to recognise a certain triangle as a right-angled one. In Q.16, this was easier because a triangle with sides $3, 4$ and 5 is the most familiar example of a right angled triangle with integer sides. In the present problem, the recognition comes only after rewriting $2\sqrt{2}$ as $\sqrt{8}$.

Q.27 Two ships are are approaching a port along straight routes of constant velocities. Initially the two ships and the port formed an equilateral triangle. After the second ship traveled 80 km., the triangle becomes right angled.



When the first ship reaches the port, the second ship was still 120 km from the port. Find the initial distance of the ships from the port.

(A) 240 km. (B) 300 km. (C) 360 km. (D) 180 km.

Answer and Comments: (A). Let x be the initial distance between the ships and the port. To find x, we introduce an auxiliary variable. Even though each ship is traveling at a uniform speed, these two speeds are different, or else, they would reach the port simultaneously. Ship II is slower than Ship I. Let λ be the ratio of the speed of Ship II to that of Ship I. Then $\lambda < 1$.

We construct a system of two equations in these two unknowns. When Ship I covers the entire distance x, Ship II has covered only $x - 120$. Hence

$$x - 120 = \lambda x \tag{1}$$

When Ship II has traveled 80 km., it is still $x - 80$ km. away from the port. From the right angled triangle with the vertex angle $60°$, the distance between Ship I and the port is half of the distance between Ship II and the port. That means at that time Ship I has covered a distance $x - \frac{x-80}{2} = \frac{x+80}{2}$ km. This gives another equation

$$80 = \lambda \frac{x + 80}{2} \tag{2}$$

As our interest is only in x, we eliminate λ by dividing (1) by (2) to get

$$\frac{x-120}{80} = \frac{2x}{x+80} \tag{3}$$

On simplification, this gives a quadratic in x, viz.

$$x^2 - 200x - 9600 = 0 \tag{4}$$

whose roots are $100 \pm \sqrt{19600} = 100 \pm 140$. Taking the positive root, $x = 240$ km. ∎

A simple but good problem testing the ability to interpret the data correctly and to translate it into a system of equations. The trigonometric part needed is minimal, only that $\sin 30^\circ = \frac{1}{2}$, which too, can be bypassed by a standard geometric property of a $30-60-90$ degrees triangle, studied in schools. Incidentally, from either (1) or (2), we get $\lambda = \frac{1}{2}$, that means Ship I is twice as fast as Ship II. In fact, it would have been a better problem to ask the ratio of the speeds of the two ships because eliminating x between (1) and (2) is not as easy as eliminating λ. So, most candidates would find x before finding λ.

Q.28 If $x_1 > x_2 > \ldots > x_{10}$ are real numbers, what is the least possible value of

$$\left(\frac{x_1 - x_{10}}{x_1 - x_2}\right)\left(\frac{x_1 - x_{10}}{x_2 - x_3}\right)\ldots\left(\frac{x_1 - x_{10}}{x_9 - x_{10}}\right)?$$

(A) 10^{10} (B) 10^9 (C) 9^9 (D) 9^{10}

Answer and Comments: (C). The numbers x_i are not given. It is not even given if they are positive. But that does not matter. The expression depends only on the differences of the successive x_i's. Specifically, let $d_i = x_i - x_{i+1}$ for $i = 1, 2, \ldots 9$. Then $x_1 - x_{10} = d_1 + d_2 + \ldots + d_n$ and the expression to be minimised is $\dfrac{(d_1 + d_2 + \ldots + d_9)^9}{d_1 d_2 \ldots d_9}$. As all d_i's are positive, by a direct application of the A.M.-G.M. inequality, this is minimum when the d's are all equal. But in that case, the expression is $\dfrac{(9d_1)^9}{d_1^9} = 9^9$. ∎

A simple problem once the idea that each numerator is the sum of all denominators strikes.

Q.29 The range of values that the function

$$f(x) = \frac{x^2 + 2x + 4}{2x^2 + 4x + 9}$$

takes as x varies over all real numbers in the domain of f is

(A) $\frac{3}{7} < f(x) \leq \frac{1}{2}$ (B) $\frac{3}{7} \leq f(x) < \frac{1}{2}$
(C) $\frac{3}{7} < f(x) \leq \frac{4}{9}$ (D) $\frac{3}{7} \leq f(x) \leq \frac{1}{2}$

Answer and Comments: (B). Normally, finding the range of a function requires calculus methods to identify the intervals on which it is increasing/decreasing and consolidating the information gained from all such intervals. But other methods are available when the function $f(x)$ has a particularly simple form. The present problem falls in this category. In fact, several methods are available which do not use calculus or use it only marginally.

The stipulation about the domain is redundant because since $2x^2 + 4x + 9 = 2(x+1)^2 + 7$, the denominator never vanishes and so $f(x)$ is defined for all $x \in I\!R$. In fact, if we put $u = (x+1)^2$, then

$$f(x) = \frac{u+3}{2u+7} \tag{1}$$

As x varies over $I\!R$, u varies over $[0, \infty)$. So the range of $f(x)$ is the same as that of $g(u) = \frac{u+3}{2u+7}$ as $u \in [0, \infty)$. Finding the range of $g(u)$ is simpler than finding the range of $f(x)$. We can actually go further. Since $g(u)$ takes only positive values, we can take its reciprocal, say $h(u) = \frac{2u+7}{u+3}$. Then the range of $g(u)$ consists of the reciprocals of the elements in the range of $h(u)$. We rewrite $h(u)$ as

$$\begin{aligned} h(u) &= \frac{2u+7}{u+3} = \frac{2u+6+1}{u+3} \\ &= 2 + \frac{1}{u+3} \qquad (2) \end{aligned}$$

As u varies over $[0, \infty)$, $\frac{1}{u+3}$ decreases from $\frac{1}{3}$ and assumes values as near to 0 as we like but not 0. Hence the range of $\frac{1}{u+3}$ is $(0, \frac{1}{3}]$. Adding 2, the range of $h(u)$ is $(2, \frac{7}{3}]$. Hence the range of the reciprocal function $g(u)$ is $[\frac{3}{7}, \frac{1}{2})$. As noted earlier, this is also the range of $f(x)$. ∎

Notice how the range was found by gradually reducing the problem to the problem of finding the range of $\frac{1}{u+3}$ for $u \geq 0$, for which calculus was needed only to the extent that $\frac{1}{u+3} \to 0$ as $u \to \infty$.

The special feature of $f(x)$ that it is a ratio of two quadratic polynomials enables another method based solely on the criterion for a quadratic equation to have real roots. To say that a real number y is in the range of $f(x)$ is equivalent to saying that the equation

$$\frac{x^2 + 2x + 4}{2x^2 + 4x + 9} = y \tag{3}$$

has at least one real solution. For a fixed y this can be rewritten as a quadratic in x, viz.

$$(2y - 1)x^2 + (4y - 2)x + 9y - 4 = 0 \tag{4}$$

We have to exclude $y = \frac{1}{2}$ because in that case there are no solutions. For $y \neq 2$, (4) will have a real solution in x if and only if the discriminant is non-negative, i.e.

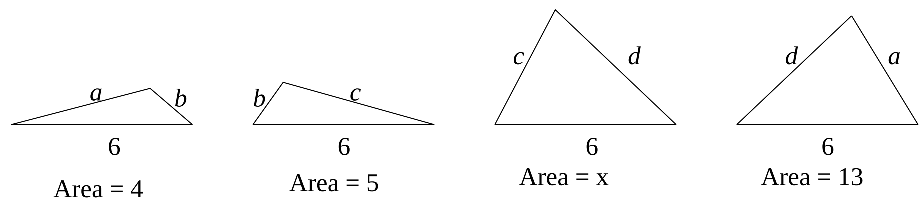
$$(2y - 1)^2 \geq (2y - 1)(9y - 4) \tag{5}$$

As we are assuming that $2y - 1 \neq 0$, there are two possibilities depending on the sign of $2y - 1$. If $2y - 1$ is positive we can cancel it from (5) to get $2y - 1 \geq 9y - 4$ or equivalently, $7y \leq 3$, i.e. $y \leq \frac{3}{7}$. But this is inconsistent with $2y - 1 > 0$. So this possibility gives no points in the range of $f(x)$. Assume, however, that $2y - 1 < 0$. Then after canceling it from (5), the inequality is reversed and gives

$$2y - 1 \leq 9y - 4 \tag{6}$$

or equivalently, $y \geq \frac{3}{7}$. Combining this with $2y < 1$, we get that (3) has a real solution if and only if $\frac{3}{7} \leq y < \frac{1}{2}$. So the range of $f(x)$ is the interval $[\frac{3}{7}, \frac{1}{2})$.

Q.30 In the following diagram, four triangles and their sides are given. Areas of three of these are also given. Find the area of the remaining triangle.



(A) 12 (B) 13 (C) 14 (D) 15

Answer and Comments: (C). The problem is ill-posed. We can express the areas of the triangles in terms of their sides using Heron's formula mentioned (but not used) in the solution to Q.26. That gives a system of three equations in the four unknowns a, b, c, d. For example, the first triangle gives

$$(a+b+6)(a+b-6)(a+6-b)(b+6-a) = (4S)^2 = 256 \qquad (1)$$

The second and the fourth triangle will give similar equations. If we can find the values of c and d from these, Heron's formula will give the area of the third triangle.

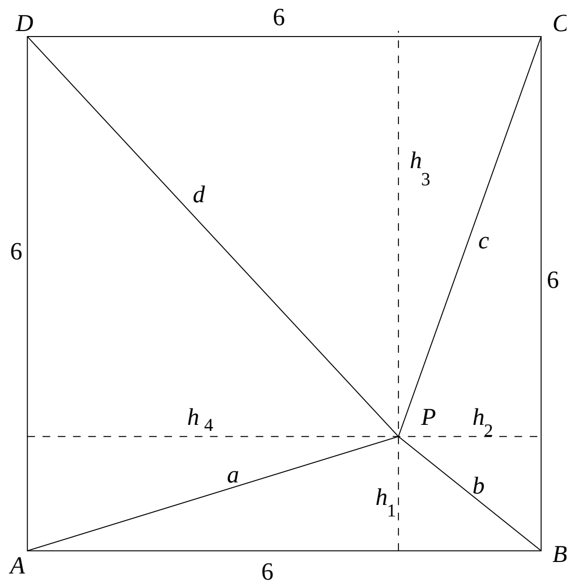
But in general, a system of three equations in four unknowns has infinitely many solutions. In the solution to Q.24, we remarked that if the variables are to take only non-negative integers as values, then sometimes the solution set becomes finite and in some cases the solution is unique too. But here the problem does not specify that a, b, c, d are integers. Nor does it follow from the data.

Nevertheless we can explicitly obtain one particular solution of this system of three equations by a simple geometric construction. All the four given triangles have 6 as one side. Let h_1, h_2, h_3, h_4 be the altitudes of the triangles falling on this side. Then the known values of the areas of the first, the second and the last triangle imply three equations, viz.

$$h_1 = \frac{4}{3}, \quad h_2 = \frac{5}{3} \text{ and } \quad h_4 = \frac{13}{3} \qquad (2)$$

The value of h_3 is not given. If we know it we can find x immediately.

If we express the altitudes in terms of sides, the system of equations we shall get will be the same as the one above (of which only one equation was shown explicitly in (1)). But the advantage of the present system is that a particular solution can be constructed very easily. We first construct a square $ABCD$ of side 6 and mark a point P inside it which is at a perpendicular distance h_1 from AB, h_2 from BC and h_4 from AD as shown below. (This is feasible since $h_2 + h_4 = 6$ which is the perpendicular between the sides AD and BC.)



If we call PA, PB, PC and PD as a, b, c, d respectively, then the triangles APB, BPC, CPD and DPA are congruent to the four given triangles. So the third triangle CPD has altitude $h_3 = 6 - h_1 = 6 - \frac{4}{3} = \frac{14}{3}$. Therefore its area x equals $\frac{1}{2} \times 6 \times \frac{14}{3} = 14$. (Alternately, x can be found by subtracting from 36 the sum of the areas of the other triangles.) ∎

By Pythagoras theorem, the sides come out as $a = \sqrt{h_1^2 + h_4^2} = \frac{\sqrt{185}}{3}, b = \frac{\sqrt{41}}{3}, c = \frac{\sqrt{221}}{3}$ and $d = \frac{\sqrt{365}}{3}$. It is doubtful if anybody can get, or even guess, these by solving the system consisting of (1) and two similar equations. And, even if someone can do that, unless it is shown that this is the only solution, we cannot conclusively prove that $x = 14$. All we can say is that 14 is a possible value of x. Sometimes solutions are unique in degenerate cases. For example the single equation $x^2 + y^2 = c$ in two variables x and y has infinitely many solutions for $c > 0$. But for $c = 0$, there is only one solution, viz. $x = 0, y = 0$. It is unlikely that in the present problem, because of some degeneracy in the data, the solution $a = \sqrt{h_1^2 + h_4^2} = \frac{\sqrt{185}}{3}, b = \frac{\sqrt{41}}{3}, c = \frac{\sqrt{221}}{3}$ and $d = \frac{\sqrt{365}}{3}$ is the only solution.

Another possibility is that sometimes the nature of the problem is such that despite the plurality of solutions, the answer is unique. For example, suppose we are asked to find $3x + 3y$ given that $x + y = 7$. This single equation does not determine the values of x and y. But that does not matter as far as the answer is concerned, which is 21. If this scenario is to happen in the present problem, then it must be shown that regardless of which solution of the system of three equations in the four unknowns a, b, c, d we take, the value of x remains the same. But that is also unlikely to happen.

Those familiar with jigsaw puzzles may think that the square in the second figure is obtained by putting together the four triangular pieces given in the first figure. But that is not so. On the contrary, we started with the square in the second figure and partitioned it suitably into triangles which satisfy the data in the problem. If we start with some other quadruple of triangles in the first diagram, we may not be able to assemble them into a square. Let α be the angle between the sides 6 and b of the first triangle. Then its area is $3b \sin \alpha$. As this equals 4 we get $3b \sin \alpha = 4$. Similarly, if β is the angle between the sides 6 and b in the second triangle, we would get $3b \sin \beta = 5$. But if these two angles are to be abutted along the common side b so as to form a corner of a square, α and β must be complementary angles, i.e. $\sin \beta$ should equal $\cos \alpha$. Although this happens to be the case in the solution above (because $\tan \angle ABP = \frac{h_1}{h_2} = \frac{4}{5}$), it may not be the case with every quadruple of triangles in the first figure.

ISI BStat-BMath-UGA-2022 Paper 2 with Comments

Q.1 Consider a board having 2 rows and n columns. Thus there are $2n$ cells in the board. Each cell is to be filled in by 0 or 1.

(a) In how many ways can this be done such that each row sum and each column sum is even ?

(b) In how many ways can this be done such that each row sum and each column sum is odd ?

Solution and Comments: (a) 2^{n-1}, (b) 2^{n-1} if n is even and 0 if n is odd.

Let S be the set $\{1, 2, \ldots, n\}$. Every placement of 0's and 1's in the cells of the first row determines a unique subset A of S, consisting of those i's for which the i-th cell in the first row has 1 in it. Similarly, every filling of the sells in the second row determines a subset B of S defined by $\{i : i\text{-th cell in the second row is filled with } 1\}$. For example, suppose we have $n = 9$ and the following filling of the cells.

1	2	3	4	5	6	7	8	9
0	1	0	0	1	1	1	0	1
0	0	1	0	1	0	1	0	1

Then $A = \{2, 5, 6, 7, 9\}$ and $B = \{3, 5, 7, 9\}$. These two subsets determine the placement completely.

Now in each column there are two entries. The column sum is even when both of them are 1 or both are 0. So, if it is the i-th column, then either i is in both A and B or in neither of them. If every column sum is to be even, then for every $i = 1, 2, \ldots, n$, either both A and B contain i or neither contains i. This is equivalent to saying that $A = B$. So, in (a), the sets A and B should be equal and moreover each should have an even number of elements, since the row sum is also to be even. The number of subsets of S having an even number of elements is the binomial sum $\displaystyle\sum_{r=0}^{k} \binom{n}{2r}$ where the upper index k of the summation is $\frac{n}{2}$ if n is even and $\frac{n-1}{2}$ if n is odd. It is well known that this sum equals

2^{n-1}, i.e. half the total number of subsets of S. (See Equation (24) in Chapter 5.) So this is the answer to (a).

For (b), both A and B must have an odd number of elements since each row sum is odd. As for column sums, for the i-th column, for the sum to be odd, i is in exactly one of A and B. That means B is the complement of A w.r.t. the set S. Summing up, the answer to (b) is the number of subsets A of S for which both A and $S - A$ have odd number of elements. If n is odd there is no such subset. If n is even, then any subset of even cardinality has this property and the number of such subsets is 2^{n-1} as already noted. So the answer to (b) is 0 if n is odd and 2^{n-1} if n is even. ∎

A simple problem once the idea of relating filling of cells with a pair of subsets of S strikes. The binomial identity needed is well-known.

Q.2 Consider the function

$$f(x) = \sum_{k=1}^{m}(x-k)^4,\ x \in I\!R,$$

where m is an integer. Show that f has a unique minimum and find the point where the minimum is attained.

Solution and Comments: The case $m = 1$ is trivial for in this case $f(x) = (x-1)^4$ is minimum at the point 1. More generally, for any $a \in I\!R$, $(x-a)^4$ attains its unique minimum at a.

When $m = 2$, $f(x) = (x-1)^4 + (x-2)^4$. It is symmetric about the midpoint $\frac{3}{2}$ of $[1, 2]$. $f(x)$ is differentiable everywhere and $f'(x) = 4[(x-1)^3 + (x-2)^3] = 4[(x-1)^3 - (2-x)^3]$ which vanishes only when $x-1 = 2-x$, i.e. at $x = \frac{3}{2}$. $f(x)$ changes its behaviour from decreasing to increasing at $\frac{3}{2}$. Hence there is a unique minimum at $\frac{3}{2}$.

Before dealing with higher values of m, we note that the argument for $m = 2$ can be generalised if $f(x)$ were of the form $(x-a)^4 + (x-b)^4$ for any two real numbers a, b, with $a < b$. There would be a unique minimum at the midpoint $\frac{a+b}{2}$.

Now suppose m is any integer bigger than 2. First suppose m is even, say $m = 2k$. We express $f(x)$ as the sum of k functions, by grouping together two terms of $f(x)$ which are symmetric about the midpoint $\frac{m+1}{2}$. That is, we let $f_1(x) = (x-1)^4 + (x-m)^4, f_2(x) = (x-2)^4 + (x-m+1)^4, \ldots, f_k(x) = (x-k)^4 + (x-(k+1))^4$. Then $f(x) = f_1(x) + f_2(x) + \ldots + f_k(x)$. In general, little can be said about the location of the minimum of the sum of several functions in terms of the minima of those functions. But in the present case, each $f_i(x)$ attains its minimum at the same point, viz. $\frac{m+1}{2}$. Hence so does their sum.

When m is odd, say $m = 2k+1$, the point of symmetry is $k+1$. we group together $(x-1)^4$ with $(x-m)^2$, $(x-2)^4$ with $(x-m+1)^4$ and so on. Each group attains its minimum at $\frac{m+1}{2} = k+1$. This time $(x-k-1)^4$ is left out. But that also attains its minimum at $k+1$. Hence $f(x)$ attains its minimum at $x = k+1 = \frac{1+m}{2}$.

Thus we have shown that for every m, $f(x)$ attains its unique minimum at $\frac{m+1}{2}$. ∎

A simple problem once the idea of expressing $f(x)$ as a sum of functions all having their minima at the same point strikes. Unlike the cases $m = 1, 2$, the general case cannot be generalised if $f(x)$ were of the form $\sum_{k=1}^{m} (x-a_k)^4$ where $a_1, a_2, \ldots, a_m$ are arbitrary real numbers.

Q.3 Consider the parabola $C : y^2 = 4x$ and the straight line $L : y = x + 2$. Let P be a variable point on L. Draw the two tangents from P to C and let Q_1 and Q_2 denote the two points of contact on C. Let Q be the mid-point of the line segment joining Q_1 and Q_2. Find the locus of Q as P moves along L.

Solution and Comments: The elaborate description of Q could have been shortened. The segment Q_1Q_2 is called the chord of contact of the point P w.r.t. the parabola C. So, Q is simply the midpoint of the chord of contact of P w.r.t. C.

Formulas are available that give the equation of the chord of contact of a point (h, k) w.r.t. any conic. Here the conic is the parabola $y^2 = 4x$ and the equation of Q_1Q_2 is

$$ky = 2(x+h) \tag{1}$$

To determine the points of contact $Q_1 = (x_1, y_1)$ and $Q_2 = (x_2, y_2)$ (say) we solve this simultaneously with the equation of C, i.e.

$$y^2 = 4x \tag{2}$$

Eliminating x, we have a quadratic in y, viz.

$$y^2 = 2ky - 4h \tag{3}$$

The roots, say y_1 and y_2 can be found. But as we are interested only in the midpoint of Q_1Q_2, we want only $\frac{y_1+y_2}{2}$ which is simply k even without solving (3). To find the x-coordinate of the midpoint, we simply put $y = k$ in (1) to get $x = \frac{k^2-2h}{2}$.

Since the point $P = (h, k)$ lies on the line $y = x + 2$ to begin with we have $k = h + 2$. Therefore the locus of Q in parametric form is

$$\begin{aligned} y &= h + 2 \\ \text{and} \quad x &= \frac{(h+2)^2 - 2h}{2} = \frac{h^2 + 2h + 4}{2} \end{aligned} \tag{4}$$

The question does not specifically ask the locus in a non-parametric form. It comes by eliminating the parameter h from these two equations. Thus the locus of Q is $2x = (y-2)^2 + 2(y-2) + 1 = y^2 - 2y + 1 = (y-1)^2$. (Note that this is also a parabola, which has the same axis and vertex as the original parabola.) ∎

An absolutely routine problem, hardly suitable for Paper 2. Ready-made formulas are also available for the mid-point of the chord of contact. But it is preferable to get it from (1) which is easier to remember both as the equation of the tangent (in case the point is on the conic) and that of the chord of contact. It is a special case of the equation of the chord of contact of a point (x_1, y_1) for a conic with equation $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ which is $axx_1 + byy_1 + h(xy_1 + yx_1) + g(x + x_1) + f(y + y_1) + c = 0$ and popularly remembered as $T = S_1$. Of course, one can proceed directly from the basic principles as in the answer to Problem 15 above. But that amounts to duplicating the proofs of the standard formulas. The best compromise is to rely on $T = S_1$, applicable for any conic.

Q.4 Let $P(x)$ be an odd degree polynomial in x with real coefficients. Show that the equation $P(P(x)) = 0$ has at least as many *distinct* real roots as the equation $P(x) = 0$.

Solution and Comments: Every polynomial with real coefficients has a real root. In fact, as a function from $I\!R$ to $I\!R$, it is surjective. That is, for every $y_0 \in I\!R$, there is at least one $x_0 \in I\!R$ such that $P(x_0) = y_0$. (This follows because $P(x) - y_0$ is also a polynomial of odd degree.)

Now, assume that the distinct roots of $P(x)$ are $y_1, y_2, \ldots, y_m$. Then for every $i = 1, 2, \ldots, m$, there is some $x_i \in I\!R$ such that $P(x_i) = y_i$. Then $P(P(x_i)) = P(y_i) = 0$. Further, for $ineqj$, $x_i \neq x_j$ as otherwise we would have $y_i = P(x_i) = P(x_j) = y_j$, contradicting that the y's are distinct.

Hence $P(P(x)) = 0$ has at least m distinct roots. ∎

Another simple question. While the existence of a root of a real polynomial of odd degree is fairly well-known, the fact that the function defined by it is onto is not used so frequently. That is the only thought needed here.

Q.5 For a positive integer n and $i = 1, 2$, let $f_i(n)$ denote the number of divisors of n of the form $3k + i$ (including 1 and n). Define for any positive integer n,

$$f(n) = f_1(n) - f_2(n).$$

Find the values of $f(5^{2022})$ and $f(21^{2022})$.

Solution and Comments: Every odd number is either of the form $3k + 1$ for some k or of the form $3k + 2$ for some k. Both 5^{2022} and 21^{2022} are odd and hence so are all their divisors. The problem asks to count how many are of the two given types.

In terms of residue classes, every odd number is congruent modulo 3 to either 1 or 2. The latter is the same as the residue class of -1, which is more convenient in taking powers.

As 5 is a prime, the only divisors of 5^{2022} are of the form 5^r where r can be any integer from 0 to 2022. Since 5 is congruent modulo 3 to

-1, 5^r is congruent to 1 modulo 3 if r is even and to -1 if r is odd. As r is to lie between 0 and 2022, $f_1(5^{2022})$ is the number of even integers between 0 to 2022. Its count is 1012 since $r = 0$ is also to be included. So, $f_1(5^{2022}) = 1012$. Contributions to $f_2(5^{2022})$ come from odd values of the exponent. They number $2023 - 1012 = 1011$. Hence

$$f(5^{2022}) = f_1(5^{2022}) - f_2(5^{2022}) = 1012 - 1011 = 1 \tag{1}$$

The method for $f(21^{2022})$ is basically the same. But 21 is not a prime. It factors as the product of two primes, viz. 3 and 7. So every divisor of it is of the form $3^r 7^s$ where r, s are integers taking values from 0 to 2022. Since the residue class of 7 modulo 3 is 1, all powers of 7 are congruent to 1 modulo 3. Hence the residue class modulo 3 of $3^r 7^s$ is the same as that of 3^r. This behaves like 5^r since, like 5, 3 is also congruent to -1 modulo 2. Therefore $f_1(21^{2022})$ is the number of ordered pairs of the form (r, s) where both r and s range over 0 to 2022 and r is even. Hence

$$f_1(21^{2022}) = 1012 \times 2023 \tag{2}$$

and similarly,

$$f_2(21^{2022}) = 1011 \times 2023 \tag{3}$$

Subtracting,

$$f(21^{2022}) = 2023. \tag{4}$$

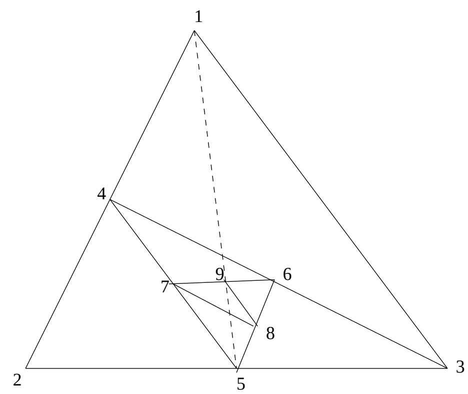
So the answer is $f(5^{2022}) = 1$ and $f(21^{2022}) = 2023$. ∎

Another simple problem based on residue arithmetic. There is little point in asking it after Problem 20 in Paper 1, which involves a little more work.

Q.6 Consider a sequence $P_1, P_2, \ldots$ of points in the plane such that P_1, P_2, P_3 are non-collinear and for every $n \geq 4$, P_n is the midpoint of the line segment joining P_{n-2} and P_{n-3}. Let L denote the line segment joining P_1 and P_5. Prove the following:

(a) The area of the triangle formed by the points P_n, P_{n-1}, P_{n-2} tends to 0 as n goes to ∞.

(b) The point P_9 lies on L.



Solution and Comments: (a) is easy. Every time we take a new P_n, it divides a side of the earlier triangle with vertices $P_{n-1}, P_{n-2}, P_{n-3}$. Hence the area of $\triangle P_n P_{n-1} P_{n-2}$ is half the area of $\triangle P_{n-1} P_{n-2} P_{n-3}$. Therefore the areas form a geometric progression with common ratio $\frac{1}{2}$. Since $(\frac{1}{2})^n \to 0$ as $n \to \infty$, (a) follows. (See the figure above, where to avoid cluttering, P_i is shown merely as i.)

For (b), we label the points P_i's with complex numbers. The conclusion is independent on the choice of the frame of reference and scaling. So, without loss of generality, we take $P_1 = 0$, $P_2 = 1$ and $P_3 = z$, some (unknown) complex number. Then the complex numbers representing P_4 to P_9 can be calculated recursively from their construction. Thus

$$P_1 = 0 \tag{1}$$
$$P_2 = 1 \tag{2}$$
$$P_3 = z \tag{3}$$
$$P_4 = \frac{0+1}{2} = \frac{1}{2} \tag{4}$$
$$P_5 = \frac{1+z}{2} \tag{5}$$
$$P_6 = \frac{z+\frac{1}{2}}{2} = \frac{2z+1}{4} \tag{6}$$
$$P_7 = \frac{z+2}{4} \tag{7}$$

$$P_8 = \frac{4z+3}{4} \tag{8}$$
$$\text{and} \quad P_9 = \frac{3z+3}{8} \tag{9}$$

The line segment L joins 0 and $\frac{z+1}{2}$. Clearly $\frac{3z+3}{8}$ lies on it since $\frac{3z+3}{8}$ is a positive real multiple of $\frac{z+1}{2}$. Hence P_9 lies on L. In fact, it divides L in the ratio 3 : 1. ▮

Another very simple problem not suitable for Paper 2. (a) is trivial as soon as it is recognised that every time the area is divided by 2. (b) could have been asked in Paper 1, asking to find the least $n > 5$ such that P_n lies on L.

Q.7 Let
$$P(x) = 1 + 2x + 7x^2 + 13x^3,\ x \in \mathbb{R}.$$
Calculate for all $x \in \mathbb{R}$,
$$\lim_{n\to\infty} \left(P(\frac{x}{n})\right)^n.$$

Solution and Comments: By a direct calculation,

$$P(\frac{x}{n}) = 1 + \frac{2x}{n} + \frac{7x^2}{n^2} + \frac{13x^3}{n^3} \tag{1}$$

If $x = 0$, then $P(\frac{x}{n}) = 1$ and the sequence $(P(\frac{x}{n}))^n$ is the constant sequence 1. Hence the limit in question is 1.

Assume $x \neq 0$. Put $h = \frac{x}{n}$. Then

$$P(\frac{x}{n}) = 1 + 2h + 7h^2 + 13h^3 \tag{2}$$

and so,

$$\left(P(\frac{x}{n})\right)^n = (1 + 2h + 7h^2 + 13h^3)^{x/h} \tag{3}$$

As $n \to \infty$, $h \to 0$ only through the values of the form $x, \frac{x}{2}, \frac{x}{3}, \ldots$. But the R.H.S. of (3) makes sense for all h in a sufficiently small deleted neighbourhood of 0 (so small that the expression $1+2h+7h^2+13h^3$ is positive). Hence the limit, say L in the question can also be evaluated if $\lim\limits_{h\to 0}(1+2h+7h^2+13h^3)^{x/h}$ can be evaluated. This limit is in an 1^∞ indeterminate form. So, we take the log of the expression and see if it has a limit.

$$\ln\left((1+2h+7h^2+13h^3)^{x/h}\right) = x\frac{\ln(1+2h+7h^2+13h^3)}{h} \tag{4}$$

x is a constant. The limit of the other factor is simply the derivative (w.r.t. h) of $\ln(1+2h+7h^2+13h^3)$ at $h=0$. By a direct calculation, it equals $\frac{2+14h+39h^3}{1+2h+7h^2+13h^3}\Big|_{h=0} = 2$. Hence

$$\lim_{h\to 0}(1+2h+7h^2+13h^3)^{x/h} = 2x \tag{5}$$

Taking exponentials, the limit of the R.H.S. of (3) and hence also the limit asked is 2^{2x}. ∎

Another routine problem for those used to evaluate limits in indeterminate form.

Q.8 Find the minimum value of

$$|\sin x + \cos x + \tan x + \cot x + \sec x + \mathrm{cosec} x|$$

for real numbers x not multiple of $\pi/2$.

Solution and Comments: This problem is a straight replica of Q.27 in Paper 1 of ISI Entrance Test 2014, except for the order of the terms in the expression for the function and the stipulation that multiples of $\pi/2$ are to be excluded from the domain of the function. The answer is $2\sqrt{2}-1$. See the commentary for that year for the solution. The change of order of the terms and the exclusion of multiples of $\pi/2$ were justified in the commentary. They have been implemented this year. But such an outright duplication is shocking to say the least.

Q.9 Find the smallest positive real number k such that the following inequality holds

$$|z_1+z_2+\ldots+z_n| \geq \frac{1}{k}(|z_1|+|z_2|+\ldots+|z_n|).$$

for every positive integer $n \geq 2$ and every choice of $z_1, z_2, \ldots, z_n$ of complex numbers with non-negative real and imaginary parts.

[Hint: First find k that works for $n = 2$. Then show that the same k works for any $k \geq 2$.]

Answer and Comments: $k = \sqrt{2}$. The wording of the question is somewhat clumsy. It would have been better to ask for the greatest positive real number M such that

$$|z_1 + z_2 + \ldots + z_n| \geq M(|z_1| + |z_2| + \ldots + |z_n|) \tag{1}$$

holds for all $z_1, z_2, \ldots, z_n$ lying in the first quadrant. Once such M is found, k is merely its reciprocal.

Following the hint, we first work out the case $n = 2$. Now the problem is to find the largest real number M such that

$$|z_1 + z_2| \geq M(|z_1| + |z_2|) \tag{2}$$

for all z_1, z_2 in the first quadrant. When z_1, z_2 are both 0, equality holds for any M no matter how large. We must therefore exclude this case because we want M to work in all cases. When at least one of z_1 and z_2 is non-zero, $|z_1| + |z_2| > 0$ and therefore (2) is equivalent to

$$\frac{|z_1 + z_2|}{|z_1| + |z_2|} \geq M \tag{3}$$

Call the L.H.S. as $f(z_1, z_2)$. It is a real valued function of two complex variables defined over the set, say $S = \{(z_1, z_2) \in \mathbb{C} \times \mathbb{C} : (z_1, z_2) \neq (0, 0)\}$. The number M we are looking for is the minimum of $f(z_1, z_2)$ as (z_1, z_2) ranges over the subset, say $T = \{(z_1, z_2) \in \mathbb{C}^+ \times \mathbb{C}^+) : (z_1, z_2) \neq (0, 0)\}$ where $\mathbb{C}^+$ denotes the set of complex numbers in the first quadrant, i.e. whose real and imaginary parts are non-negative.

On the entire set S, the minimum of f is 0. It occurs when $z_1 + z_2 = 0$. But such pairs (z_1, z_2) are not in the subset T. So, it is possible that the minimum of f on the subset T is higher. (Note that when a function is restricted to a subset, its maximum decreases but its minimum increases in general.)

Let us analyse when f assumes smaller values. If we view z_1, z_2 as vectors and the angle between them is obtuse, then their components along the angle bisector between them are oppositely directed and serve to nullify each other. The larger the angle between them, the larger the extent to which they cancel each other. In the extreme case, when z_1 and z_2 are oppositely directed, the angle between them is 180 degrees, the degree of cancellation is highest and we get $|z_1 + z_2| = ||z_1| - |z_2||$. At the other extreme, when the angle between z_1 and z_2 is 0 degrees they add up and we get $|z_1 + z_2| = |z_1| + |z_2|$.

This suggests that to minimise $f(z_1, z_2)$ over the subset T, we should choose the angle between z_1 and z_2 as large as possible subject to the constraint that both are in the first quadrant. The largest angle possible with this restriction is a right angle. For example, if we take $z_1 = 1$ and $z_2 = i$, then the ratio $\dfrac{|z_1 + z_2|}{|z_1| + |z_2|}$ equals $\dfrac{|1 + i|}{1 + |i|} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$. Hence the minimum of $f(z_1, z_2)$ over T is at least $\frac{1}{\sqrt{2}}$. Since $\frac{1}{\sqrt{2}}$ is one of the values assumed by f on T, the minimum M of f on T is at least $\frac{1}{\sqrt{2}}$. We claim, however, that M is actually the minimum of f on T, that is, we show that for any $(z_1, z_2) \neq (0, 0)$ with z_1, z_2 lying in the first quadrant,

$$f(z_1, z_2) = \frac{|z_1 + z_2|}{|z_1| + |z_2|} \geq \frac{1}{\sqrt{2}} \tag{4}$$

This is trivially true if z_1 or z_2 is 0. So assume that both are non-zero. Write them in the polar form as $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ where both θ_1, θ_2 are in $[0, \frac{\pi}{2}]$. Without loss of generality assume that $\theta_2 \leq \theta_1$. Let $z = z_1/z_2$. Then $z = re^{i\theta}$ where $r = \frac{r_1}{r_2}$ and $\theta = \theta_1 - \theta_2$. Then z also lies in the first quadrant. Dividing both the numerator and the denominator by $|z_2|$,

$$f(z_1, z_2) = f(z, 1) = \frac{|z + 1|}{|z| + 1} \tag{5}$$

So, (4) is equivalent to proving that $2|z + 1|^2 \geq (|z| + 1)^2$. Since $|z + 1|^2 = |z|^2 + 1 + z + \overline{z} = 1 + r^2 + 2r\cos\theta$ we are reduced to proving that $2 + 2r^2 + 4r\cos\theta \geq 1 + 2r + r^2$. But that is true because $1 - 2r + r^2 = (1 - r)^2$ and $\cos\theta$ are both non-negative (since z is in the first quadrant).

Thus we have proved (2) with $M = \frac{1}{\sqrt{2}}$. Hence $k = \sqrt{2}$ works for $n = 2$. This completes the first part of the hint given. To extend the inequality for any $n \geq 3$, first assume that $n = 3$. Then z_1, z_2, z_3 are linearly dependent. Without loss of generality, assume z_3 lies in the acute angle between z_1 and z_2. Then $z_3 = \alpha z_1 + \beta z_2$ where α, β are non-negative. Hence $1 + \alpha$ and $1 + \beta$ are positive. Using the result in the case $n = 2$, we have

$$
\begin{aligned}
|z_1 + z_2 + z_3| &= |(1+\alpha)z_1 + (1+\beta)z_2| \\
&\geq \frac{1}{\sqrt{2}}[|(1+\alpha)z_1| + |(1+\beta)z_2|] \\
&= \frac{1}{\sqrt{2}}[(1+\alpha)|z_1| + (1+\beta)|z_2|] \\
&= \frac{1}{\sqrt{2}}[|z_1 + |z_2| + (|\alpha z_1| + |\beta z_2|)] \\
&\geq \frac{1}{\sqrt{2}}[|z_1| + |z_2| + |z_3|] \qquad (6)
\end{aligned}
$$

where in the last step we have used the triangle inequality $|\alpha z_1| + |\beta z_2| \geq |\alpha z_1 + \beta z_2|$.

The proof for the case $n > 3$ is similar. Without loss of generality we assume that $z_3, z_4, \ldots, z_n$ all lie in the acute angle between z_1 and z_2. (To express this more precisely, suppose z_1 has the least and z_2 the greatest argument among $z_1, z_2, \ldots, z_n$.) For $i = 3, 4, \ldots, n$, write z_i as $\alpha_i z_1 + \beta_i z_2$ where α_i and β_i are non-negative. Then $z_1 + z_2 + \ldots + z_n = w_1 + w_2$ where $w_1 = (1 + \alpha_3 + \alpha_4 + \ldots + \alpha_n)z_1$ and $w_2 = (1 + \beta_3 + \beta_4 + \ldots + \beta_n)z_2$. Applying the result for the case $n = 2$ to the pair (w_1, w_2), regrouping the terms and finally applying the triangle inequality for $\alpha_i z_1 + \beta_i z_2$ for each $i = 3, 4, \ldots, n$ establishes the result for n.

So $M = \frac{1}{\sqrt{2}}$ and the number k asked is $\sqrt{2}$. ∎

An elementary but excellent, thought provoking problem on absolute values of complex numbers. We are thoroughly familiar with the triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$. In a multiplicative form, 1 is an upper bound on the ratio $\dfrac{|z_1 + z_2|}{|z_1 + z_2|}$. We know when equality holds. We also know when the upper bound is way too high. The present problem deals with a situation where the upper bound is not so high by restricting the z_1 and z_2 to a subset of the complex plane. The hint given is very helpful because the arguments needed for $n = 2$ and for extending the result to $n > 2$ are very different. The extension hinges crucially on the linear dependence of three (or more) complex numbers. If the result is paraphrased in terms of the triangle inequality for vectors, then an extension from $\mathbb{R}^2$ to $\mathbb{R}^3$ will not be valid since three vectors in $\mathbb{R}^3$, even though given to lie in the first orthant, may be linearly independent.

CONCLUDING REMARKS

The test this year is highly disappointing, especially on the backdrop of the last year. Paper 2 is especially lacking in any imaginatively designed problems. The only exception is Q.9. Q.8 is also an excellent problem, but a straight repetition of a 2014 problem. Q.2 in Paper 2 is a good problem because of the short cut provided by the symmetry of the function to be minimised. Solution of Q.1 is based on a certain binomial identity, which, although fairly well-known, is not among the basic identities. In a full length question, this leads to a controversy whether a candidate is expected to prove that identity.

The only (moderately) noteworthy problems in Paper 1 are :

1. Problem 4: About a polynomial $P(x)$ satisfying $|P(x)| \leq |x|$ for all real x.
2. Problem 5: About the functional equation $f(x)f(y) = f(x+y)+f(x-y)$.
3. Problem 11: Maximisation of $12xy(72 - 3x - 4y)$.
4. Problem 12: Minimisation of $|x-3| + |x+1| + |x+2| + |x|$
5. Problem 13: About a function whose average over any interval occurs at the midpoint..
6. Problem 16: About a triangle inscribed in a square.
7. Problem 19: About magic squares of order 3
8. Problem 20: Divisors of $n^5 + 4n^4 + 3n^3 + 2022$.
9. Problem 24: About the integral solutions of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 + \frac{2}{abc}$.
10. Problem 26: Finding the in-radius of a triangle with sides given.

Out of these, Problem 16 and 26 test the ability to simplify work involved by noticing some special features of the data. Problem 24 is not likely to be solved without a tricky conversion of the given equation.

There are a few other good problems in Paper 1. But they lack novelty as similar problems have been asked several times in comparable entrance tests.

Problem 30 of Paper 1 seems ill posed since it asks to determine solutions of a system of three equations in four unknowns. Analogy with jigsaw puzzles is tempting but misleading.

As usual, there is duplication of certain ideas while omission of some areas. For example, both Problem 11 and 28 of Paper 1 are based on the A.M.-G.M. inequality. Although number theory is represented in three problems, they are not upto the mark. Probability is paid a lip service. There are no problems on conditional probability. And, of course none on the integral part function. But the most shocking omission is differential equations.

The best problem in the entire test is the last problem of Paper 2 about the triangle inequality for complex numbers.